



Team 15 Sprint 3 Planning Document

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Sprint Overview

This sprint is primarily focused on implementing social features in *Forge*, such as uploading workouts and diets, and supporting various user interactions. Along with that, we will be working to integrate previously made features, such as workout and meal logging, to work with the new social features to make all features collaborate seamlessly. Finally, we will be fixing last-minute issues and improving features where we can to present as high-quality a final product as we can.

SCRUM Master: Drew Balaji

Meeting Plan: 3:00 to 4:00 PM Thursday and Sunday at Hicks Undergraduate Library

Risks and Challenges:

The risks are the actual uploading of workouts and diets in real time to display on the user feed. Syncing uploads by users with their friends' feed will be difficult, and even messaging and liking for users is a complex task. Coordinating our efforts to tackle these issues will be key, as if one of these aspects is implemented improperly, the whole social network that we build could break completely.

Current Sprint Detail

User Story 34

As a gym user, I would like to be able to upload my workouts to a community space.

Task	Description	Estimated Time	Member Assigned
1	Create pop up to upload a workout after clicking log workout	2 Hours	Luke
2	Add button to upload a workout that is already logged by a user	1 Hour	Luke
3	Develop logic to move a workout from the database to be displayed of the community feed	3 Hours	Luke
4	Insert endpoint, connect frontend buttons with backend retrieval of uploaded workouts	2 Hours	Luke
5	Testing to ensure that workouts are uploaded to the community space properly	2 Hours	Luke

Acceptance Criteria:

- Assuming the upload workout pop-up works on the add workout screen, if the user presses the log workout button, the user will be prompted to upload the workout to a feed.
- If the upload workout button works, if the user selects to upload the workout, they should be able to see it displayed in the social feed.
- If another user uploads a workout, if another user is friends with that user, they should see the newly uploaded workout quickly

User Story 35

As a social user, I would like to be able to share or show off my progress to my friends.

Task	Description	Estimated Time	Member Assigned
1	Determine how progress will be displayed to friends (workout progression, weight loss, etc.)	2 Hours	Luke
2	Create frontend-backend logic to pull values from the database and generate relevant values to show for user progress	3 Hours	Luke

3	Add interactable buttons that display a progression graph for other users' progress they have shared	2.5 Hours	Jack
4	Testing that shared progress is accurate, and friends see the accurate progress	2 Hours	Luke

Acceptance Criteria:

- Assuming the user does not have any/enough data to show progression, if the user's profile is selected and the progression tab is selected, then no relevant values should be shown for the statistics.
- Assuming the user has enough data to show progression, if the progression tab of a user is selected, relevant and accurate information about the user should be displayed
- Assuming the user is blocked/unfriended, if the blocked user selects the blockee's profile, then the blocked user should not be able to see their progression
- If the user selects a display graph button, the appropriate progression and future projection should be displayed on a graph

User Story 36

As a user, I will be able to see a feed of posts from my friends

Task	Description	Estimated Time	Member Assigned
1	Use friend relationships from 42 to determine which post should appear in the friends feed	2 Hours	Artem
2	Create backend to retrieve posts from friends	3 Hours	Artem
3	Frontend UI for a friends feed tab	3 Hours	Artem
4	Connect backend to frontend	2 Hours	Artem
5	Tests to ensure proper functionality	2 Hours	Artem

Acceptance Criteria:

- If a user opens the friends feed, posts from their current friends will display
- If a user is not friends from one someone or else has another user block, their posts will not display
- If a friend uploads a new post and the feed is refreshed, the new post should appear

User Story 37

As a user, I will be able to see a feed of posts from my gym

Task	Description	Estimated Time	Member Assigned
1	Determine which posts should appear on the feed	2 Hours	Artem
2	Create backend to retrieve posts from other users in the same gym	3 Hours	Artem
3	Frontend UI for a gyms feed tab	3 Hours	Artem
4	Connect backend to frontend	2 Hours	Artem
5	Testing to ensure proper functionality	2 Hours	Artem

Acceptance Criteria:

- If a user opens the gym feed, posts from other users in the same gym should appear
- If a post belongs to a user from a different gym, that post should not appear in the gym feed
- If a user from a selected gym uploads a post, and the feed is refreshed, that post should appear

User Story 38

As a social user, I would like to be able to view my friends' streaks.

Task	Description	Estimated Time	Member Assigned
1	Develop UI streak indicators in the profiles for friends	3 Hours	Pratik
2	Implement backend logic to calculate the streak of a user based on the current data found in their account	3 Hours	Pratik
3	Notify users on a loss of a streak, or a new streak starting	2.5 Hours	Pratik
4	Testing to ensure that streaks correspond to correct data, and ensure that users are able to see streaks	1.5 Hours	Pratik

Acceptance Criteria:

- If streaks are implemented properly, if a user has been in the gym for a week, a one should appear as the user's streak, and for each week that number should be incremented

- If friends viewing streaks works properly, if a friend views the profile streak of a user, they should see the same value that the user themselves sees for their account
- If letting users know when they lose/gain streaks is implemented, if a user breaks a streak they had started, they will be notified in the app and encouraged to start a new one
- If users are told when they start streaks properly, if a user logs a workout for the first time since their last streak, they will be told that their new streak has started

User Story 39

As a social user, I would like to be able to message other users in the app.

Number	Description	Estimated Time	Member Assigned
1	Determine which users can message each other (friends only, blocked users, etc)	2 Hours	Artem
2	Update database to store messages between users	4 Hours	Artem
3	Create backend to to send and receive message	3 Hours	Pratik
4	Back frontend UI to display messages	3 Hours	Pratik
5	Connect frontend to backend	2.5 Hours	Pratik
6	Testing to ensure proper functionality	2 Hours	Pratik

Acceptance Criteria:

- If a user opens the messaging page, they should be able to view their existing conversations.
- If a user selects a conversation, the messages should appear in chronological order.
- If a user receives a message from another user, that message should display

User Story 40

As a dieter, I would like to be able to upload custom meals to a community space (tied in with 41)

Number	Description	Estimated Time	Member Assigned
1	Implement logic allow users to send a created meal to the community page	4 Hours	Drew
2	UI button in diet tab	2.5 Hours	Drew
3	UI to let all of the community meals show	3 Hours	Drew

	up		
4	Developer Test: share a meal to feed with api/server call, verify with database read	2 Hours	Roan

Acceptance Criteria:

- If a user clicks the share button, the meal will appear in the social tab
- Users will be prompted with a confirmation to send the meal to the social tab
- If I change my mind about adding my meal, I will be able to delete the meal from the community space

User Story 41

As a dieter, I would like to be able to save meals from the community space (tied in with 40)

Number	Description	Estimated Time	Member Assigned
1	Implement logic to let users “download” meals from the social tab to their diet tab	4 hrs	Drew
2	UI button in social tab	2.5 hrs	Drew
3	UI updates in diet tab	3 hrs	Drew
4	Developer Test: search through meals in feed, call api/server to add to log, verify with database read	2 hrs	Roan

Acceptance Criteria:

- If a user clicks the download button, the meal will appear in the diet tab
- Users will be prompted with a confirmation to send the meal to the diet tab
- If I change my mind about adding my meal, I will be able to delete the meal from my meal browser

User Story 42

As a user, I will be able to add and remove friends from my social network

Number	Description	Estimated Time	Member Assigned
1	UI capabilities to search all users by username and add/remove them as friends	2 Hours	Roan
2	Backend workflows to query and display users	3 Hours	Roan

3	Update database schema to track friend relationships	2 Hours	Roan
4	Backend workflows to add and remove friends	3 Hours	Roan
5	Developer Test: add a friend then remove them with api/server calls	1 Hour	Roan

Acceptance Criteria:

- User can see the desired user displayed after searching username
- User can add the friend to their network and see confirmation
- User can remove the friend from their network and see confirmation

User Story 43

As a social user, I would like to be able to “like”, react, and comment on different shared workouts and meals.

Number	Description	Estimated Time	Member Assigned
1	Implement comment structure system inside of database, alongside	3 Hours	Jack
2	Implement buttons on posts that allow users to like and react, and shows likes/reactions	2 Hours	Jack
3	Implement comment feed that shows previous users comments chronologically	2 Hours	Jack
4	Implement comment input that allows users to comment on other posts	3 Hours	Jack
5	Implement manual tests for social interactions	1 Hour	Jack

Acceptance Criteria:

- If a user views a post, they can see the like and reaction count.
- If a user clicks the comment feed button on a post, a clean scrollable embedding of previous comments will be displayed.
- If the user clicks a like or reaction button on a post, the posts like and reaction counts will update accordingly.
- If a user clicks to write a comment, the comment will be uploaded and appear in the post's comment embedding.

User Story 44

As a user, I will be able to view a Notifications page to view friend requests and likes & comments on my posts

Number	Description	Estimated Time	Member Assigned
1	Implement a UI button that allows the user to view a notification feed	2 Hours	Jack
2	Implement a notification feed that shows a scrollable list of the user's notifications, such as likes, comments, and friend requests	4 Hours	Jack
3	Implement a notification object such that interactable notifications like accepting/denying friend requests are differentiated from static notifications such as likes and comments	3 Hours	Jack
4	Interactable notifications should perform tasks on the database, such as adding friends or removing themselves	2 Hours	Jack
5	Write manual tests	1 Hour	Jack

Acceptance Criteria:

- If a user clicks the notification button, they should see a notifications feed
- If a user interacts with a friend request notification, the user should see the friend added or the friend request denied, based on their choice.
- If a user closes the notification feed, it should return them to the previous page

User Story 45

As a user, I will be able to block and/or report other users

Number	Description	Estimated Time	Member Assigned
	Reuse user search & display from 42		
1	UI capabilities to block a user and/or report a user with a reason	2 Hours	Roan
2	Update database schema to track blocked relationships	2 Hours	Roan
3	Backend workflows to block/unblock users	3 Hours	Roan
4	Developer Test: block a user then unblock them with api/server calls	1 Hour	Roan

Acceptance Criteria:

- User can see the desired user displayed after searching username
- User can report them with a description of the violation
- User can block them personally from sharing & viewing posts

User Story 46

As a social user, I would like to be able to set collaborative goals with my friends, where we all work toward similar goals and hold each other accountable.

Number	Description	Estimated Time	Member Assigned
1	Logic to allow users to first create a goal and then invite friends to participate in the goal	7 Hours	Drew
2	Create button UI which will go to another popup allowing the user to input their goal	4 Hours	Drew
3	Make the goals pop up in the social tab and allow friends to see the goal and progress	4 Hours	Drew
4	Logic to alert friends that they've been added to a goal and the goal has been reached	3 Hours	Drew
5	Testing to ensure that collaborative goals with friends are able to be completed together as intended	1 Hour	Pratik

Acceptance Criteria:

- Users are presented with a create button to create a group goal
- Friends of users can see the created goal and chart their progress; users can also uninvite themselves from the goal
- Once a goal is completed, users will be alerted to the goal being finished.

User Story 47

As a user, I will be able to attach my gym location to my profile

Number	Description	Estimated Time	Member Assigned
1	Update UI capabilities on the settings page with a drop down of nearby gyms	2 Hours	Roan
2	Update database schema to track discrete gym locations	2 Hours	Roan

3	Backend workflows to support the functionality	2 Hours	Roan
4	Developer Test: read database, select gym and write to database, reread database	1 Hour	Roan

Acceptance Criteria:

- User recognizes that their current gym location is unknown as a baseline
- User can select a gym from a list of nearby gyms and save to profile
- User can back out of settings then reopen to see the updated gym location

User Story 48

As a social user, I would like to be able to decide whether my progress is public or available only to specific users.

Number	Description	Estimated Time	Member Assigned
1	Develop UI to allow a user to select between public and restricted access	2 Hours	Pratik
2	Develop a difference in screens for whether a user viewing the account has or does not have access to view progress	3 Hours	Pratik
3	Add logic that on entering an account, determine if user is public or restricted and if the entering user has access	3 Hours	Pratik
4	Testing to ensure full functionality of public and restricted access for all sorts of users	1.5 Hours	Pratik

Acceptance Criteria:

- If public status is implemented properly, if a user is set to be public, any valid user can see their progress
- If restricted status is implemented properly, if a user is not set to public, only selected users/friends are able to see the progress for the non-public user
- If public status is implemented properly, if a public user has blocked a certain account, that blocked user should not be able to see the public user's progress.

Sprint 2 - Leftover User Stories and Tasks

User Story 14

As a user, I would like to be able to select a location, so I can see gyms and people near me.

Number	Description	Estimated Time	Member Assigned
1	Synchronize with Maps API to embed a display of nearby gym locations	2 Hours	Jack
2	Query the database to connect with nearby friends that also have location access	3 Hours	Jack
3	Testing to ensure proper map functionality	1 Hour	Jack

Acceptance Criteria:

- Given that the UI button is implemented correctly, the nearby gyms menu should display to the user on click.
- Given that the UI for requesting user permissions is implemented correctly, the user should receive a pop-up for location access upon attempting to view nearby gyms.
- Given that the Maps API is properly implemented, the user should be able to view an embedded map with nearby gyms pinned.
- Given that the Maps API and database are properly implemented, the user should be able to view nearby friends and users.

User Story 22

As a gym user, I will be able to generate an all-inclusive, tailored, generic Quick Workout, accounting for every possible profile characteristic.

Number	Description	Estimated Time	Member Assigned
1	Allow input from the user to change aspects of a generated workout	3 Hours	Luke
2	Log workouts in the database when users select to log the generated workout	2 Hours	Luke
3	Testing proper logging and editing of workouts from the users works as anticipated	1 Hours	Luke

Acceptance Criteria:

- The user can manually make any changes they desire
- Assuming the Quick Workout feature and the database are interconnected, if the user selects the log workout button, it should be logged for the user correctly.

User Story 23

As a gym user, I will be able to generate a tailored workout purely based on a specific muscle group.

Number	Description	Estimated Time	Member Assigned
1	Ensure that workout generation errors are generated for invalid inputs	0.5 Hours	Luke
2	Allow users to select multiple muscles to have a more targeted workout	1.5 Hours	Luke
3	Testing to ensure that workout generation still works and not inputting a muscle returns an error	1 Hour	Luke

Acceptance Criteria:

- If the error handling is done properly, if the user generates a workout for a muscle group without explicitly saying a muscle group, the user should be prompted with an error message, and no workout should be generated.
- If the multiple muscle group select logic works, if a user selects multiple muscles, the workout returned should have a workout that targets each of those muscles

User Story 29

As a user, I would like to be able to have weekly and monthly diet and workout reports generated.

Task	Description	Estimated Time	Member Assigned
1	Calculate statistics based on user data	2.5 Hours	Luke
2	Connect reports to the frontend, displaying proper values for the user	2 Hours	Luke
3	Testing to ensure that for different data users are generated the proper weekly/monthly report	1 Hour	Luke

Acceptance Criteria:

- Given that the reports are generated properly, if a report is generated for a specific user, the report should reflect the data for that user specifically, and not any other user in the database.

Remaining Backlog

Functional Requirements:

General

- ~~As a user, I will partake in an onboarding process that will consist of age, height, weight, goals, and experience~~
- ~~As a user, I would like to be able to create and manage an account for the app, such as creating and changing a username or bio.~~
- ~~As a user, I would like to be able to reset my password.~~
- ~~As a user, I would like to be able to easily navigate to the workout, diet, and social tabs through the use of a menu.~~
- ~~As a user, I would like to be able to read and agree to the Terms and Conditions.~~
- ~~As a user, I would like to be able to delete my account.~~
- ~~As a user, I would like to be able to set up notifications for workouts, meals, etc.~~
- ~~As a user, I would like to be able to enable certain accessibility features (dark mode, large text).~~
- ~~As a user, I would like to be able to maintain a week-by-week workout.~~
- As a user, I would like to be able to maintain diet streaks.
- ~~As a user, I would like to be easily able to convert recipe and weight amounts between metric and imperial.~~
- ~~As a user, I would like to be able to select a location, so I can see gyms and people near me.~~
- ~~As a user, I will be able to bypass manually inputting username and password during login with a Remember Me feature~~
- As a user, I would like to be able to select a location, so I can see gyms and people near me.
- ~~As a user, I would like to be able to enable certain accessibility features (dark mode, large text).~~
- As a user, I will be able to export my profile to show a recap of long-term progress

Workout

- ~~As a gym user, I would like to be able to track my workouts by logging the workout, reps, weight, and sets.~~
- ~~As a gym user, I would like to be able to have cardio machines suggested to me for targeted fat loss.~~
- ~~As a gym user, I would be able to upload and save my prior gym routines.~~

- As a gym user, I would like to be able to upload my workouts to a community space.
- ~~As a gym user, I would like to be able to add tags to my workouts to make them easier to find.~~
- ~~As a gym user, I would like to be able to search for different workouts based on the tags assigned to them.~~
- As a gym user, I would like to be able to view shared workouts and do them myself.
- ~~As a gym user, I would like to be able to schedule my workouts on a calendar.~~
- As a beginner on my fitness journey, I would like to know where to start and how to make consistent progress.
- As an experienced athlete, I would be able to identify key techniques and strategies to improve that I otherwise wouldn't have noticed.

Diet

- ~~As a dieter, I would be able to view the macros of different meals from national restaurants.~~
 - This can be provided through a large database such as [MenuStat](#).
- As a dieter, I would like to be able to chart my daily macro and water intake, and I would like to be able to set macro and water goals to hit daily.
- As a dieter, I would like to be able to set certain junk foods to avoid.
- ~~As a dieter, I will press a button to recalibrate my caloric goals based on recorded behavior.~~
- ~~As a dieter, I will press a button to recalibrate my caloric goals based on recorded behavior.~~
- ~~As a new user, I will have my day-one caloric goals displayed based on onboarding responses.~~
- ~~As a dieter, I would like to be able to apply certain tags to my meals to make them easier to find.~~
- ~~As a dieter, I would like to be able to search for different meals using the tags assigned to them.~~
- As a dieter, I would like to filter community meals by servings, calories, protein, cuisine, complexity, and other important components.
- As a user, I would be able to assign macros and ingredients to my meals when I add them.
- As a dieter, I would like to be able to react to different meals using a like feature, emojis, or comments.
- As a dieter, I would like to select ingredients I currently have to create meals from them.
- As a dieter, I would like help settling into a diet with less strict requirements initially, or a cheat day to allow myself an easier transition.
- As a dieter, I would like to know what national chain meals fit into my diet.

Social

- As a dieter, I would like to be able to upload custom meals to a community space.
- As a dieter, I would like to be able to save meals from the community space.
- As a social user, I would like to find other weightlifters nearby.
- As a social user, I would like to be able to “friend” other users.
- As a social user, I would like to be able to message other users in the app.
- As a social user, I would like to be able to “unfriend” any users.
- As a social user, I would like to be able to block any users who harass me.
- As a social user, I would like to be able to report any users who violate the app TOC.
- As a social user, I would like to be able to share or show off my progress to my friends.
- As a social user, I would like to be able to decide whether my progress is public or available only to specific users.
- As a social user, I would like to be able to view my friends’ streaks.
- As a social user, I would like to be able to “like” and comment on different shared workouts and meals.
- As a social user, I would like to be able to set collaborative goals with my friends, where we all work toward similar goals and hold each other accountable.

AI

- As a user, I would like to be able to have weekly and monthly diet and workout reports generated.
- ~~As a gym user, I would like to be able to ask questions to get clarifications on specifics, such as the best number of reps or sets for a certain exercise.~~
- ~~As a gym user, I would like to be able to have alternative options suggested if a machine or exercise isn’t available or is undesirable.~~
- As a gym user, I would like to have workouts change based on the inability to do certain workouts, such as due to injury.
- As a gym user, I would like to achieve a natural weight progression based on how often I work out on a specific exercise.
- As a gym user, I would like advice on how to recover from an injury.
- As a gym user, I would be able to take a video of my lifts and harness AI to correct my form if need be.
 - Due to the large scale of this story, this will take significantly longer than other stories, approximately 30 hours. This was discussed with the coordinator.
 - The primary challenges in this story come from being able to analyze form using AI, as it would require a large dataset of exercise videos to train on to compare an arbitrary video. It would not only have to recognize the technique but compare it to a “perfect” technique as well, and pass the results to an LLM to formulate feedback.

- ~~As a gym user, I would like exercises recommended to me based on my experience level, which I have stated in my onboarding.~~
- As a gym user, I would like to be able to generate a gym routine based on the muscle group to train.
- ~~As a dieter, I would be able to have recipes suggested to me for various goals related to weight loss or muscle gain.~~
- ~~As a beginner in my fitness journey, I would like tips or advice on how to do a specific exercise for the first time, embedded as a help button.~~